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AQA A Level Further Mathematics (7367)

Topic C: Matrices

Sub topic C3: Matrix Transformations (3D)

Mark Scheme

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Mark Scheme

Topic **C: Matrices**

Sub topic **C3: Matrix Transformations (3D)**

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To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

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Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Identifies correct values of sine **and** cosine.	AO1.1a	M1	$\cos\theta = -\frac{1}{2}$ and $\sin\theta = \frac{\sqrt{3}}{2}$
Selects correct angle. Accept $\frac{2\pi}{3}$ or $-240°$ or $-\frac{4\pi}{3}$	AO1.1b	A1	$\theta = 120°$
Deduces the transformation giving a full description. FT their angle. Accept $\frac{2\pi}{3}$ or $-240°$ or $-\frac{4\pi}{3}$ Condone missing degree sign. Condone missing 'anticlockwise'. NMS scores 3/3	AO2.2a	A1F	Rotation about the z-axis through 120° anti-clockwise.
			Total 3 marks

Q2.

(a) (i) Reflection

Reflection stated for M1

M1

In (the plane) $z = 0$ (or in the $x - y$ plane)

Either version for A1

A1

2

(ii) Rotation

Rotation stated.

M1

About the y-axis

y-axis

A1

through $\frac{\pi}{3}$ radians

(or 60°)

B1

3

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(b) $T_1T_2 = \begin{bmatrix} \frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \\ 0 & 1 & 0 \\ \frac{-\sqrt{3}}{2} & 0 & \frac{1}{2} \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 & \frac{-\sqrt{3}}{2} \\ 0 & 1 & 0 \\ \frac{-\sqrt{3}}{2} & 0 & -\frac{1}{2} \end{bmatrix}$

M1 correct order of matrices

A1 fully correct

[N.B $\begin{bmatrix} \frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{3}}{2} & 0 & -\frac{1}{2} \end{bmatrix}$ *scores M0A0]*

M1A1

2

(c) (i) For line of invariants points

$\begin{bmatrix} k & 2 & -1 \\ 1 & 1 & 1 \\ 3 & 4 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

Set up equations − uses $\mathbf{M}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

M1

$\Rightarrow kx + 2y - z = x \Rightarrow (k - 1)x + 2y - z = 0$ ①

$x + y + z = y \Rightarrow x + z = 0$ ②

Two equations correct

A1

$3x + 4y + z = z \Rightarrow 3x + 4y = 0$ ③

All three equations correct

A1

From ② $z = -x$

From ③ $y = \frac{-3x}{4}$

Defines variables in terms of one letter or 2 components in $\begin{bmatrix} 4 \\ -3 \\ -4 \end{bmatrix}$ *correct*

M1

Substitute in ① $(k - 1)x - \frac{3}{2}x + x = 0$

$x\left[k - 1 - \frac{3}{2} + 1\right] = 0$

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$x\left[k - \frac{3}{2}\right] = 0$

Substitution into other equations or $\begin{bmatrix} 4 \\ -3 \\ -4 \end{bmatrix}$ *correct*

A1

$x \neq 0 \quad \Rightarrow \quad k = \frac{3}{2}$

k -value obtained.

A1

(ii) Line $x = \frac{-4}{3}y = -z$

$\frac{x}{4} = \frac{y}{-3} = \frac{z}{-4}$

Or equivalent

B1cao

7

Alternative:

$\begin{bmatrix} k-1 & 2 & -1 & 0 \\ 1 & 0 & 1 & 0 \\ 3 & 4 & 0 & 0 \end{bmatrix}$

Substitute and set up

(M1)

$r_1 \to r_1 + r_2$

Row operation

(A1)

$\begin{bmatrix} k & 2 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 3 & 4 & 0 & 0 \end{bmatrix}$

$r_1 \to r_1 - \frac{1}{2}r_3$

Row operation

(A1)

$\begin{bmatrix} k-\frac{3}{2} & 2 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 3 & 4 & 0 & 0 \end{bmatrix} \Rightarrow k = \frac{3}{2}$

k-value obtained

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(A1)

$v = \lambda \begin{pmatrix} 4 \\ -3 \\ -4 \end{pmatrix}$

M1 obtains v in terms of single vector.
A1 correct

(M1A1)

$\frac{x}{4} = \frac{y}{-3} = \frac{z}{-4}$

Correct form

(B1cao) **(7)**

[14]

Q3.

(a) Reflection in $x = z$

M1 A1 **2**

(b) Rotation about the y-axis

M1 A1

Through $\cos^{-1} 0.6$ ($\approx 53.13°$)

Ignore direction

A1 **3**

[5]

Q4.

(a) (i) $\mathbf{r} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix}$

Any correct vector line equation; B1 if one vector correct, or if both correct but equation not in correct form

B2,1 **2**

(ii) $\mathbf{r} = \begin{bmatrix} 1 & 4 & -3 \\ 2 & -1 & 0 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1+2\lambda \\ 2+3\lambda \\ 3+6\lambda \end{bmatrix} = \begin{bmatrix} -4\lambda \\ \lambda \\ -\lambda \end{bmatrix}$

Attempt at multiplication

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M1

At least one entry correct

A1

All three correct

A1

Clear and valid explanation that this is a line through *O*

E1

4

(b) (i) $\begin{bmatrix}1 & 4\\ 2 & -1\end{bmatrix}\begin{bmatrix}p\\ \frac{1}{2}p+k\end{bmatrix}=\begin{bmatrix}3p+4k\\ \frac{3}{2}p-k\end{bmatrix}$

For LHS

B1

For RHS

M1A1

Answer satisfies $y = \frac{1}{2}x - 3k$

A1

4

(ii) Equal gradients, hence parallel

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ft if previous answer is of the form $y = \frac{1}{2}x + c$

E1F

Distance = $|k - c| \cos\theta$ with $\tan\theta = \frac{1}{2}$

Allow incorrect value of c here

M1

...= $\frac{8k}{\sqrt{5}}$

Allow *3.58k*

A1

3

[13]

Q5.

(a) (i) $\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

B2 **2**

(ii) $\mathbf{B} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

B1 **1**

(b) (i) $\mathbf{R} = \mathbf{BA} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Product correct way around

M1

Most correct; all correct ft ft

A1A1 **3**

(ii) Reflection in $x = 0$ (or y–z plane)

*M for correct **R***

M1A1

<u>Note 1:</u> For $\mathbf{R} = \mathbf{AB} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

*If all correct, ft their **A**, **B***

(B1)

Reflection in $y = 0$ (or x–z plane)

*Full ft, M for correct **R***

(M1)(A1)

<u>Note 2:</u> 90° rotation in –ve sense gives

$\mathbf{B} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

***A** as before*

(B1)

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$$\mathbf{R} = \mathbf{BA} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(M1)(A1)(A1)

Reflection in $y = 0$ (or x–z plane)

(M1)

Full ft (incl. Note 1 possibility –
Reflection in x = 0 (or y–z plane))

(A1) 2

[8]

Q6.

(a) Rotation

M1

about the y-axis

Ignore direction

A1

through $\cos^{-1} 0.8$

or $\sin^{-1} 0.6$ *or 36.87° or* 0.644^c

A1 3

(b) Reflection in $y = x$

Ignore if it is called a line

M1A1 2

[5]

Q7.

(a) (i) det **W** = 0

B1

Transformed shapes have zero volume

Or equivalent statement
ft volume statements

B1 2

(ii) Char eqn is $\lambda^3 - 4\lambda^2 + 4\lambda = 0$

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One A mark for each coefficient (not the λ^3)

M1A3

Solving the cubic eqn

M1

λ = 0, 2, (2)

A1

6

Alternative:

Det $(\mathbf{W} - \lambda\mathbf{I}) = \begin{vmatrix} 3-\lambda & -1 & 1 \\ 2 & -\lambda & 2 \\ -1 & 1 & 1-\lambda \end{vmatrix}$

$= \begin{vmatrix} 2-\lambda & 0 & 2-\lambda \\ 2 & -\lambda & 2 \\ -1 & 1 & 1-\lambda \end{vmatrix}$

Use of R/C ops. $R_1' \to R_1 + R_3$

M1

$= (2-\lambda)\begin{vmatrix} 1 & 0 & 1 \\ 2 & -\lambda & 2 \\ -1 & 1 & 1-\lambda \end{vmatrix}$

Factor of (2 – λ) correctly extracted

A1

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$= (2-\lambda)\begin{vmatrix} 1 & 0 & 0 \\ 2 & -\lambda & 0 \\ -1 & 1 & 2-\lambda \end{vmatrix}$

$C_1' \to C_3 - C_1$

$= (2-\lambda)^2\begin{vmatrix} 1 & 0 & 0 \\ 2 & -\lambda & 0 \\ -1 & 1 & 1 \end{vmatrix}$

A1A1

$= (2-\lambda)^2(-\lambda)$

Complete factorisation attempt

M1

giving eigenvalues 0 and 2 (twice)

A1

(6)

(b) (i) $x - y + z = 0$

B1

1

(ii) $$\begin{bmatrix} 3 & -1 & 1 \\ 2 & 0 & 2 \\ -1 & 1 & 1 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 3a - b + c \\ 2a + 2c \\ -a + b + c \end{bmatrix}$$

M1A1

$x' - y' + z' = 3a - b + c - 2a - 2c - a + b + c$

M1

$= 0 \Rightarrow P'$ in H also

Shown carefully

A1

4

[13]

Q8.

(a) A is a Rotation thro' 90°

M1 A1

about Ox

A1

B is a Reflection in $y = 0$ (i.e. $x - z$ plane)

M1 A1

5

(b) (i) $\mathbf{M}_C = \mathbf{M}_B\,\mathbf{M}_A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

M1 A1

2

(ii) C is a Reflection in $y = z$

Give M1 for any series of reflections

M1

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N.B. In (i): $\mathbf{M}_A \mathbf{M}_B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$ scores M0
but ft "Reflection in $y = -z$" in (ii)

A1

2

[9]

Q9.

(a) (i) **AB** = a 3 × 3 matrix

M1

$= \begin{pmatrix} 3 & 2 & t+1 \\ 1 & 2 & t-1 \\ 3 & 2 & t+1 \end{pmatrix}$

At least 5 elements correct, incl. at least one from C_3

A1

All elements correct

A1

3

(ii) **BA** = a 2 × 2 matrix

M1

$= \begin{pmatrix} 2 & 2 \\ t & t+4 \end{pmatrix}$

A1

2

(b) $R_1 = R_2$ ($\Rightarrow$ det **AB** = 0)

*Or expanding and **showing** det = 0*

B1

1

(c) **BA** = $\begin{pmatrix} 2 & 2 \\ -2 & 2 \end{pmatrix} = 2\sqrt{2}\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$

M1 A1

E: enlargement s.f. $2\sqrt{2}$

B1

F: Rotation

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NB: Rotation bit may be sorted completely separately in which case marks are split 3 + 3

M1

clockwise (about O) thro' 45°

Or − 45°, 315°

A1 A1

6

[12]

Q10.

Rotation

B1

about the y−axis

Ignore direction

B1

through an angle of 53.13°

Accept $\cos^{-1}$ 0.6 etc.

B1

[3]

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Q11.

(a) det **A** = 1

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B1

1

(b) The z-axis (i.e. $x = y = 0$)

B1

1

(c) Rotation

M1

about the z-axis

ft axis or correct

A1ft

through $\cos^{-1}$ 0.28

73.7° (awrt 74°) or 1.29 rads;

$\sin^{-1}$ 0.96, $\tan^{-1}\frac{24}{7}$

A1

3

[5]

Q12.

(a) Setting det **P** $(2t - 6 + 2 - 3t + 8 - 1 = 3 - t) = 0$

M1

$\Rightarrow t = 3$

A1

2

(b) (i) $\begin{bmatrix} 6 & 0 & 0 \\ -7t-21 & 3-t & 15+5t \\ 0 & 0 & 6 \end{bmatrix}$

Multiplication attempt with ≥ 3 correct elements

M1

Any one row or column

A1

All correct

A1

3

(ii) When $t = -3$, $\mathbf{PQ} = 6\mathbf{I}$

The 6 may be implicit

B1 B1

2

(iii) $\mathbf{Q}^{-1} = \frac{1}{6}\mathbf{P} = \frac{1}{6}\begin{bmatrix} 2 & 1 & 1 \\ 1 & -3 & -2 \\ 3 & 2 & 1 \end{bmatrix}$

Allow for $\frac{1}{6}$ ***P*** *or* $\frac{1}{36}$ *(6**P**)*

B1

1

(c) Enlargement (centre O) sf 6

M1 A1

2

[10]

Q13.

(a) **A:** Reflection in $y = z$

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M1 A1

B: Reflection in $y = 0$ (x-z plane)

M1 A1

4

(b) (i) **AB** = $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$

≥ 5 entries correct

B1

All correct

B1

2

(ii) About the x-axis; through 90° *+/–; or 270°; or in radians*

B1B1

2

[8]

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Examiner reports

Q1.

Most students realised that the given matrix was a rotation, although some struggled to describe it adequately. It was evident that the general matrix provided in the formulae book had been used by almost all students, but a common error was to solve only one of the trigonometric equations. A rotation of 60° from $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$ was not uncommon. However, the majority of students answered this question well.

Q2.

The two transformations were often correctly described, although on occasions the plane $z = 0$ became 'an axis' or 'line'. A small minority of students reversed the order of multiplication of the two matrices. In part (c) an eigenvector approach rarely scored many marks due to full determinant expansion errors, incorrectly setting the determinant of $\mathbf{M}_3 = 0$ or not realising that 1 was an eigenvalue. The approach of $\mathbf{M}_3\mathbf{v} = \mathbf{v}$ was far more successful apart from minor manipulation errors.

Q3.

This turned out to be a much more straightforward and routine question than question 1 for almost all candidates. The mean mark achieved by candidates was almost 4½ out of the 5 available.

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Q4.

In part (a)(i), most candidates knew how to convert the given equations into a suitable vector equation, but some lost a mark by writing 'L = ...' rather than '$\mathbf{r}$ = ...'. Part (a)(ii) was likewise answered well but often not perfectly, the usual fault here being a failure to show clearly why the line must pass through the origin.

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Part (b) was found more difficult than part (a), even though it was in a two-dimensional context. Much confusion arose from the use of the same letters x and y to denote the coordinates of a general point before and after the transformation. Many candidates were able to carry out an appropriate matrix multiplication but were unsuccessful in their attempts to form a linear equation for the image line. In part (b)(ii), most candidates earned the first mark by comparing the gradients of the two lines, even if the equation of the second line was incorrect, but very few indeed realised that the 'distance' between the two parallel lines referred to the perpendicular distance between them.

Q5.

This was another fairly basic test of the results, given in the formulae booklet, relating to the matrices of 3-d transformations, and was mostly handled very well. Apart from a small minority who made elementary mistakes with the matrices for **A** and **B**, the most common error was in taking '**A** followed by **B**' to be represented by the product **AB** rather than **BA**. A few candidates mistakenly described the reflection as being in the $y = z$ plane when they really meant the y-z plane. In marking part (b)(ii), the error of describing $x = 0$ as a line rather than a plane was overlooked. Follow-through marks were allowed for those who had multiplied **AB** rather than **BA** but in **no** other cases.

Q6.

This was a straightforward starter to the paper as might be expected, requiring candidates only to know their way round the Booklet of Formulae and interpret the standard results given therein. Centres should note that there is no intention to require candidates on this specification to specify what is meant by positive and negative directions or anti-clockwise and clockwise senses when describing rotational 3-d transformations, so no candidates were penalised for specifying such matters. Nor were candidates penalised in this case for describing $y = x$ as a line rather than a plane in part (b). Quite a few candidates did lose a mark, however, when they failed to distinguish between 'the plane $y = x$' and 'the x-y plane'.

Q7.

Incorporating a 3 × 3 matrix inevitably leads to a higher degree of difficulty in finding the characteristic equation, and this frequently proved to be so for many candidates. Once again, however, it was the introductory remark required to explain the significance of a zero determinant that caused most difficulty. Surprisingly few candidates seemed entirely sure about it, and the most popular suggestion was that det **W** = 0 meant that volumes were unchanged. However, the majority of responses dwelt exclusively on the theme of areas, which would have been suitable only had **W** been a 2 × 2 matrix. A very few candidates produced a surprising answer, explaining that **W** was a "dimension-destroying (or crushing)" matrix. This gains no credit, unless candidates then go on to explain that 3-d shapes become 2-d ones, or possibly even lines or points, at this stage. A lot of candidates wrote incoherent and meaningless statements: vague statements such as "it is co-planar" don't carry much weight and certainly don't get the marks.

Q8.

This question relied on the use of information given in the formulae booklet, and the majority of candidates coped very well with it. Many fell at the obvious hurdle in part (b), by supposing that "A followed by B" was represented by the matrix $\mathbf{M}_A\mathbf{M}_B$, rather than $\mathbf{M}_B\mathbf{M}_A$. Candidates who failed to attempt to describe the three transformations by a single, rather than multiple (ie composite), description were not awarded marks. There was no penalty for candidates who, for instance, called a plane a line, so long as they gave the right equation.

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Q9.

The two product matrices **AB** and **BA** in part (a) of this question were very popular and the majority of candidates gained all 5 marks. The rest of the question really was poorly done on the whole, but this was the most demanding work on the paper. In part (b), many of candidates noted that det (**AB**) was zero, but failed to explain why. In part (c), most seemed to resort to guesswork, especially regarding the transformation F. A shear was, marginally, the most popular choice, with a 90° rotation close behind. The majority of candidates seemed to have no method of approach to the problem. Those that took a scalar factor out of the matrix usually contented themselves with 2, 4 or 8 as the scale factor of the enlargement E; thereby leaving themselves with a matrix they simply did not know what to do with. Those who chose "rotation" as their answer then did so from a matrix with det ≠ 1 and consequently did not gain all the marks available.

Q10.

This was a straightforward starter to the paper, requiring the use of information supplied in the Booklet of Formulae. It was very rare to find a candidate who failed to score all three of the marks available here.

Q11.

Most candidates answered this question very well, making intelligent use of the results given in the formula booklet. Once again, however, there was a significant proportion of candidates who gave an inappropriate answer, often $z = 0$, when asked for the invariant line in part (b).

Q12.

This proved to be a profitable question for many candidates. In particular, the multiplication of two 3x3 matrices was handled very competently indeed, even though algebraic terms were involved. The odd error arising here did not prevent candidates from doing good work in the later parts of the question. They were not required to check that all entries of the product matrix gave consistent values for both t and k.

Q13.

Many Candidates would have benefited greatly from a keener awareness of what is given them in the **Formulae Booklet**. This was really a very direct test of those ideas.

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